1 FIG-214 — From-scratch force disagreement diagnostic

Group: MD-FROM-SCRATCH-FAILURE-ATTRIBUTION **Evidence chain:** manifested **Status:** force disagreement as diagnostic; not calibrated uncertainty

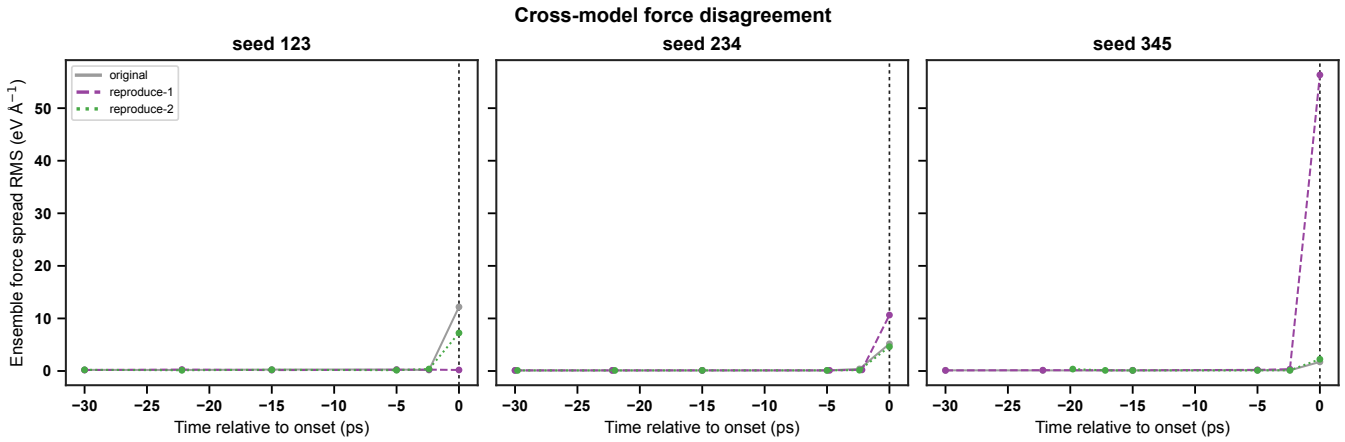


Figure 1: A–C, seed-resolved force-disagreement traces for the original and reproduction branches. The shaded or marked regions indicate the same operational instability context used in the threshold audit. The panel preserves branch- and seed-level variation; it is a diagnostic comparison and not a calibrated predictive uncertainty interval.

Results text draft. Force-disagreement traces vary across seeds and branches and are examined alongside, rather than substituted for, the operational instability onset. The figure therefore supports follow-up selection and timing comparisons without assigning a universal force-error cutoff.

Discussion and boundary. This diagnostic does not establish that force disagreement caused the trajectory failure, nor that it is calibrated uncertainty for the deployed model. Any DFT follow-up remains prepared-only in this campaign, and unstable branches remain excluded from formal transport reporting.

Article placement. Proposed article location: Supplementary Information / from-scratch instability and failure-attribution audit; not yet a main-text model-uncertainty claim.

Reviewer linkage: overall status PENDING.

- **REV-C9 (PENDING)**: Assess whether force disagreement provides actionable evidence around the from-scratch instability boundary. *Completed:* Seed-resolved traces are shown for the audited branches. *Still required:* Define the reference force target, uncertainty calibration, and prospective acceptance threshold before using this diagnostic quantitatively.

Evidence pointers.

- Data/processed: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/force_disagreement.csv`
- Data/processed: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/frame_features.csv`
- Plot/processing script: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/plot_failure_attribution_all.py`
- Plot Atlas and figure manifest: `/Users/alina/Documents/Codex/plot_library`; `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/figures/figure_manifest.json`
- Asset SHA256: `7eae1cc22b2db7d09711596482848ba7d773843e23ab43df5d7e424ee1ef9ed2`

Machine-collected evidence candidates (generated from the analysis manifest; do not edit this file).

- **Diagnostic report:** `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/RESULTS_DISCUSSION.md`

- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/diagnostic_manifest.json
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/plot_failure_attribution_all.py
- **Plot Atlas source:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: Define and validate a force-disagreement estimand and acceptance threshold prospectively before using it for quantitative model screening or uncertainty claims.